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FEATURE-GUIDED MACHINE-CENTRIC IMAGE CODING FOR DOWNSTREAM TASKS

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ABSTRACT

Video coding, a process of compressing and decompressing digital video content, has traditionally been optimized for human visual systems by reducing its size while maintaining the human perceptual quality. However, with the remarkable progress of artificial intelligence (AI) technology, the need for machine-centric coding has rapidly been increasing in recent years. In response to these trends, international standardization organizations such as MPEG are actively working to develop and launch new standards on coding technologies for machines, called video coding for machines (VCM). In this paper, we present a novel feature-guided block-wise image blending method for image compression, which is suitable for machine applications such as object detection and segmentation. For this, we use a gradient map of the feature loss using the pretrained encoder part of a task-specific network as guide for input degradation, so that the degraded input images can be effectively compressed for machine-centric tasks. Our method is simple but effective because additional training is not required by utilizing the pretrained encoder parts of networks for targeted tasks. Experimental results show that BD-rate gains can be obtained by applying our proposed method with averages 11% and 8% for object detection and instance segmentation tasks, respectively, compared to the image anchor results of MPEG-VCM reference software v0.4.

Index Terms— Coding for Machines, Image Compression, Machine vision, MPEG VCM

1. INTRODUCTION

Triggered by the emergence of AlexNet [1], deep-learning technologies for visual recognition tasks, represented by convolutional neural network (CNN), has rapidly and explosively been developed over the past decade [2-4]. For the classification task, CNNs have already surpassed human-level accuracy as early as 2015 [5], furthermore, studies on relatively higher-level visual tasks such as object detection [6-8], instance segmentation [9], object tracking [10] and others have also been intensively conducted. As the performance of neural networks for such machine vision tasks have reached a level that is applicable to real industries, considerable demands of leveraging AI technologies to various real-world applications such as autonomous vehicles, drones, smart surveillance and collaborative intelligence have been increasing in recent years [11-14]. In such applications, the primary consumer of visual data is no longer human but rather machines. In this context, researchers are increasingly paying attention to processing and compression techniques of image and video optimized for machine-to-machine data communication, called *Coding for Machines* [15-23].

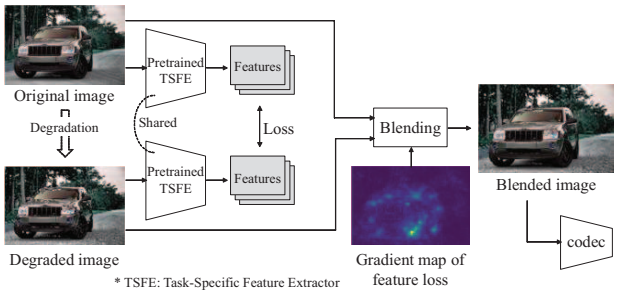

Fig. 1. Overview of the proposed scheme.

It has been proposed to improve the encoding process of conventional codecs [24-26] considering object detection and segmentation tasks [11, 16, 18, 22]. A method for improving the mode selection of encoding process [11], an efficient quantization parameters (QP) allocation scheme based on detection results of object [22] or saliency [18] were studied. Deep neural network architectures trained for machine vision-based compression have been also proposed [15, 17, 20, 23]. In [15], an image pre-transformation network was studied that can improve the classification performance. A deep generative compression framework was also presented for scalability between machine and human vision [17]. A deep compression network that can be jointly trained with key machine vision tasks such as detection, segmentation and classification was investigated [20]. H. Choi and I. V. Bajić [23] has proposed a learned image coding framework supporting latent-space scalability by constructing layered bitstreams from autoencoders. On the other hand, studies have been conducted on compression framework for collaborative intelligence environments [12-14, 19]. In [12], a compression framework for intermediate feature maps of

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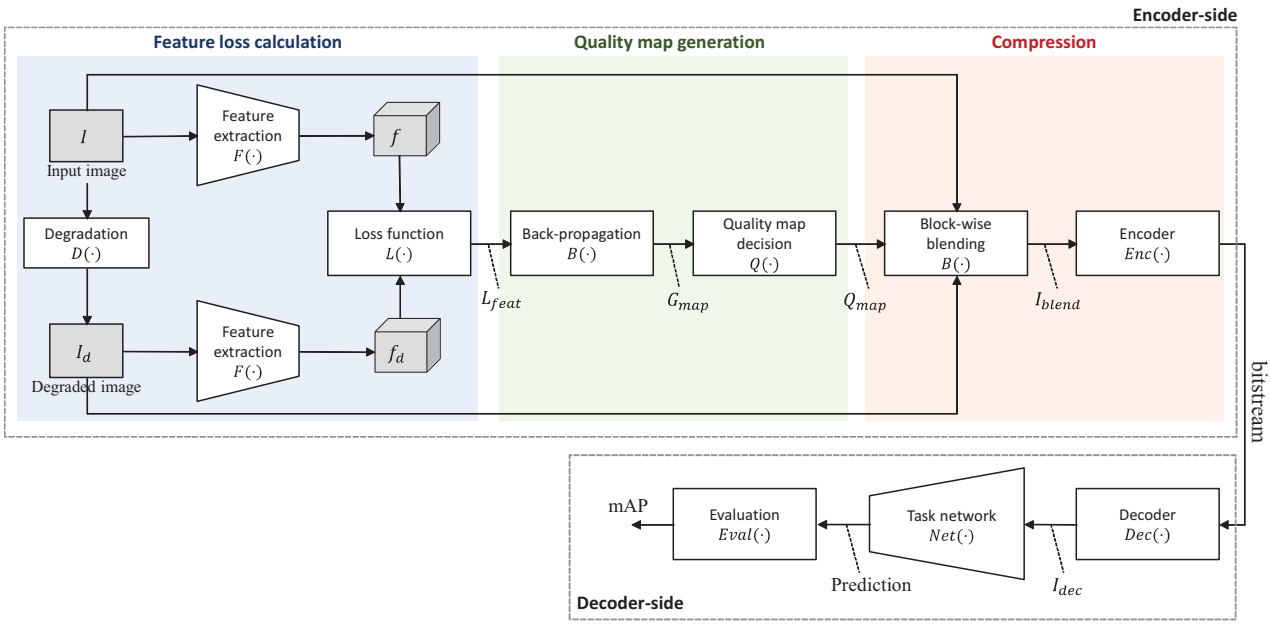

Fig. 2. Overall pipeline of the proposed feature-guided machine-centric image coding framework.

YOLO [7] was studied for collaborative object detection. A learning-based feature compression method dedicated to feature pyramid network (FPN) [8] was also proposed [19].

Besides the active research in academic fields, international standardization organizations are also paying attention to the machine-dedicated coding technologies. In particular, MPEG, which has traditionally led the evolution of video codecs by developing popular standards such as H.264/AVC [24], H.265/HEVC [25], and H.266/VVC [26], has established an Ad-Hoc group called VCM to standardize video compression techniques suitable for machine vision tasks [27]. MPEG-VCM has been exploring candidate technologies that can surpass the *anchor* generated from the latest conventional codec [26] under the common test conditions [28] and evaluation framework [29]. Currently, the standardization activities on VCM are divided into two tracks depending on the approach, one that compresses intermediate feature maps of neural networks and the other that compresses videos. Regarding the second track, core experiments are being conducted on video processing techniques such as pre-processing and post-processing suitable for machine vision.

In this paper, we propose a feature-guided image pre-processing and compression method for the machine-centric coding framework. Motivated by an idea that task performance of a machine depends on the extracted feature maps, we introduce a gradient map of a feature loss, which can be obtained by back-propagation of the difference in feature values extracted from original and degraded image. Based on the obtained gradient map, a block-wise quality map that can properly blend the original and the degraded image is obtained, and the blending is performed. In addition, experimental results of the proposed method on image datasets are compared to those of MPEG-VCM image anchor.

2. PROPOSED METHOD

The proposed feature-guided machine-centric image coding framework is shown in Figure 2. In the pipeline, the encoder-side consists of three parts which are feature loss calculation, quality map generation and compression. On the decoder-side, it is composed of decoder, task-specific neural network, and evaluation of the inference results in sequence.

2.1. Feature loss calculation

The purpose of the feature loss calculation part is to derive values that can determine relatively important regions in an input image with respect to the target machine vision task. In order to identify the region that the machine focuses on, we tried to find out which parts of the original image would cause a significant change in the machine's task performance when those are modified. Since the performance is highly related to the features extracted from a neural network of the machine, we calculated the difference in feature values between the original image and a degraded image that can approximately measure performance change. The difference is defined as a *feature loss*, and this value is used to obtain the *gradient map* through back-propagation.

First, an input image I with a size of $H \times W \times 3$ is degraded through a degradation function $D(\cdot)$ to obtain I_d, i.e., $I_d =$

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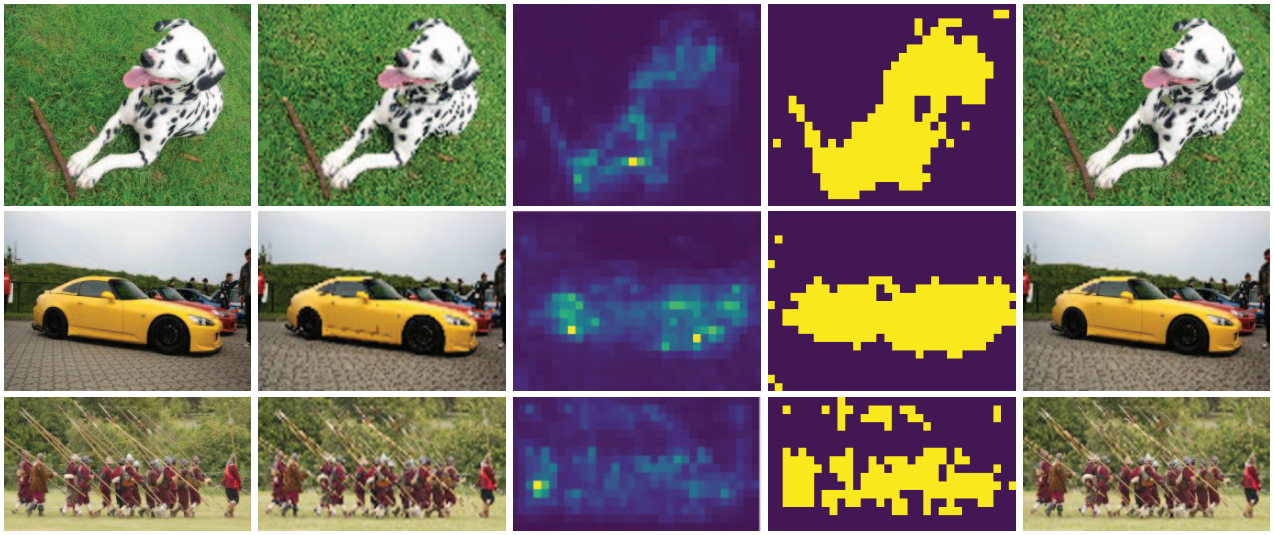

(a) Original image (b) Degraded image (c) Gradient map (d) Quality map (e) Blended image

Fig. 3. Example images for each stage of the proposed method. Note that (c) *Gradient map* is block-merged and (d) *Quality map* is binarized by a threshold for the intuitive understanding.

$D(I)$. The degradation refers to an image transformation in direction of degrading quality to obtain a compression-friendly version of the input image. Filters that reduce high-frequency components such as Gaussian blur filter can be considered, or a method of down-sampling and bi-cubic interpolating the image can also be used. Note that I_d has less quality but has the same size as I, $H \times W \times 3$. In Figure 3 (a) and (b), it shows examples of original images and corresponding degraded images.

After that, features are extracted from both images to obtain f and f_d. The same feature extraction function $F(\cdot)$ is applied to both images, which is a feature extraction part of a pre-trained task network $Net(\cdot)$. A splitting point that partitions $F(\cdot)$ from $Net(\cdot)$ can be adjusted by hyper-parameters. For instance, if the backbone of task network is Feature Pyramid Network (FPN) [8], then (a) stem, (b) C2-layer, or (c) P-layer (feature pyramid layer) can be considered as illustrated in Figure 4.

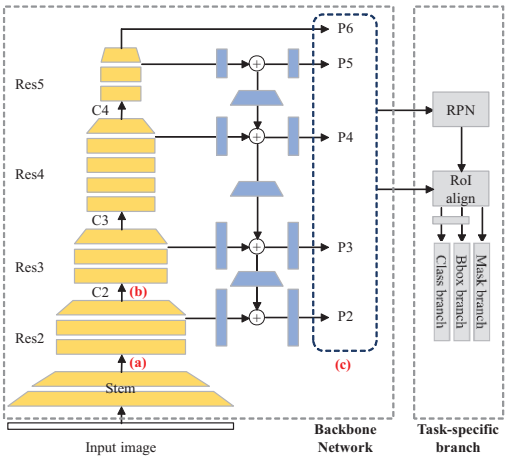

Fig. 4. Feature extraction points in Mask R-CNN with FPN backbone network: (a) stem, (b) C2 layer, (c) P-layer.

The extracted features are compared by a loss function $L(\cdot)$ generating L_{feat}. Metrics that measure the difference between the two features can be used for $L(\cdot)$, such as L1-norm, L2-norm, and cross entropy. The output of the feature loss calculation part can be described as follows.

$$L_{feat} = L\big(F(I), F(I_d)\big). \tag{1}$$

2.2. Quality map generation

In this part, a quality map for the blending process of the original image and the degraded image is generated. By back-propagating the calculated feature loss with respect to the feature extraction network $F(\cdot)$, a gradient map G_{map} can be obtained. Note that the obtained gradient map has the same spatial size as I with a single channel, i.e., $H \times W \times 1$. This map can be interpreted as implying the relative importance of each region in terms of feature values. It is because that if a certain region is activated with high values in the map, it means that changes in that region have a relatively higher impact on the variation of the loss value than others. Figure 4 (c) shows examples of the gradient map for given input images in (a).

The gradient map is then fed into quality map decision process $Q(\cdot)$ to obtain a quality map Q_{map}. This process consists of three consecutive steps, that are absolute operation,

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block-wise merging, and normalization. The absolute operation is performed to obtain the magnitude of the gradient values which is possible to be positive or negative. And these values are merged into blocks of size $N_b \times N_b$. The reason of grouping the values into a block with a certain size instead of using in pixel unit is as follows.

- First, in machine vision tasks, the relationship between values of neighboring pixels is important, not only the individual pixel value. Since CNNs extract features from an image based on sliding kernels [1-3], if there are important area in terms of machine vision, it is more valid to retain a good quality over a certain range.
- Second, from a perspective of codec, selective degradation in pixel unit is less likely to be effective than block-wise process. In conventional codecs [25, 26], an input frame is processed in a block unit called coding tree unit (CTU), so aligning the variation of the quality to borders of CTUs can have advantages for effective compression.

Note that this process reduces the size of the map into $\lceil H/N_b \rceil \times \lceil W/N_b \rceil \times 1$. The merged values are normalized to the range of [0,1], in order to have a unified blending rule regardless of characteristics of input images.

2.3. Compression

The compression part is composed of two steps which are block-wise blending of the original and degraded images, and encoding the blended image using a conventional codec. The blended image I_{blend} is generated by the following equation.

$$I_{blend}(u,v) = \begin{cases} I(u,v) & \text{if } Q_{map}\left(\left\lceil \frac{u}{N_b} \right\rceil, \left\lceil \frac{v}{N_b} \right\rceil\right) \geq Q_T, \\ I_d(u,v) & \text{otherwise,} \end{cases} \quad (2)$$

where $I_*(u,v)$ means the pixel value at coordinate (u,v) of image I_*, and Q_T is a threshold value for Q_{map} which is a hyper parameter within [0,1] and is empirically determined. Figure 4 (d) and (e) shows example images of Q_{map} thresholding by Q_T and I_{blend} blended using the binarized Q_{map}, respectively.

The blended image is encoded into bitstream by an encoder $Enc(\cdot)$ of a conventional codec such as HEVC [25] and VVC [26]. Sine I_{blend} has the same size and format as I but only differs in content, it can be directly fed into the codec with the same encoding configuration of I.

2.4. Decoder-side

The decoder-side of the proposed framework is the same as that of the MPEG-VCM anchor pipeline shown in Figure 5. The compressed bitstream is decoded by the decoder of the conventional codec, and the target machine task is then performed. The prediction results can be evaluated by task metrics such as mean average precision (mAP).

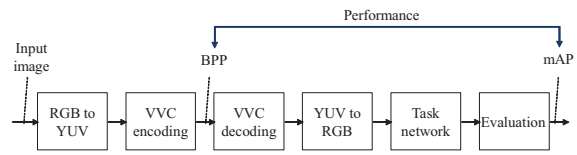

Fig.5. Image anchor generation pipeline of MPEG-VCM.

3. EXPERIMENTS

3.1. Experimental setup

The experimental setup follows the common test conditions [28] defined by the MPEG-VCM standardization activity. Two image datasets, OpenImages-5k and TVD-image, were selected for the experiments. OpenImages-5k is a subset of OpenImages V6 [30], consisting of 5,000 selected images each for object detection and instance segmentation tasks. TVD-image dataset is constructed by sampling frames from the video sequences of Tencent Video Dataset (TVD) [31]. Both datasets are labeled for object detection and instance segmentation tasks. For the codec, VVC Test Model (VTM) 12.0 [32] is used. Since the test model only supports YUV format as its input interface, format conversion between RGB and YUV was performed by ffmpeg [33] before the encoder and after the decoder. Table 1 provides more detailed information of experimental conditions and hyper parameters.

Table 1. Experimental conditions.

		OpenImages-5k	TVD-image
$D(\cdot)$		Bicubic down & upsampling	Gaussian blur
Feature extraction point		P-layer	P-layer
$L(\cdot)$		L2-norm	L2-norm
Q_T	detection	0.5	0.4
	segmentation	0.7	0.5
N_b	detection	64	64
	segmentation	32	64
Codec		VVC Test Model (VTM) 12.0 [32]	
Codec configuration		All Intra (AI) configuration	
QPs		22, 27, 32, 37, 42, 47	
Object detection network		Faster R-CNN X101-FPN (in Detectron2 [34])	
Instance segmentation network		Mask R-CNN X101-FPN (in Detectron2 [34])	
Metric		mAP@0.5	mAP@[0.5:0.95]

3.2. Results

The baseline of our experiments is the image anchor of MPEG-VCM reference software [35] which follows the pipeline in Figure 5. It compresses the input image using VVC only with format conversion, and measures the bit per

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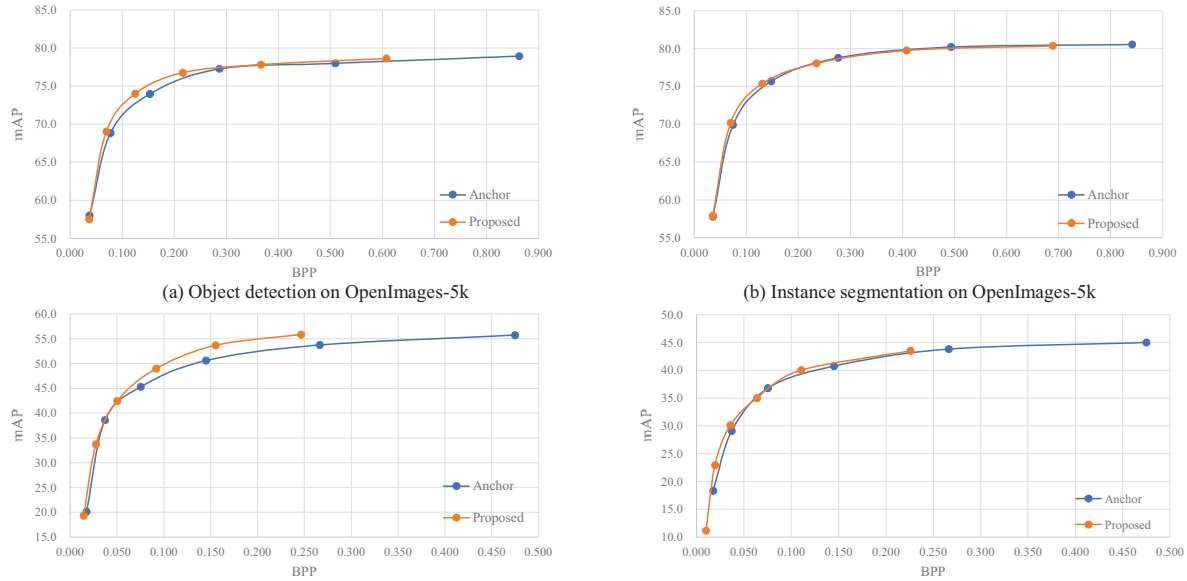

(a) Object detection on OpenImages-5k

(b) Instance segmentation on OpenImages-5k

(c) Object detection on TVD-image

(d) Instance segmentation on TVD-image

Fig. 6. Results plots: bpp vs. mAP performances of the proposed method compared to the anchor.

pixel (BPP) versus mAP. The plots of comparison results are presented in Figure 6. It can be seen that the proposed method shows improvement in performance compared to the anchor, especially for the object detection task. The BD-rate and BD-mAP gains of our proposed method from the anchor are presented in Table 2. It is confirmed that gains can be obtained for all test cases, in particular, BD-rate gains with average 11% can be acquired for the object detection task.

Table 2. BD-rate and BD-mAP gains of our method to the anchor.

	OpenImages-5k	TVD-image
	BD-rate	
Object detection	-10.82%	-12.36%
Instance segmentation	-5.92%	-10.53%
	BD-mAP	
Object detection	0.83	1.67
Instance segmentation	0.53	1.02

* BD-rate: negative is better, BD-mAP: positive is better

4. CONCLUSION

In this paper, we have presented a new coding framework for machine composed of feature-guided image blending as preprocessing and compression. The main idea of our method is to find relatively important areas of an input image in a machine-centric manner by calculating a gradient map of a feature loss. We then blend the original and degraded input images, in order to obtain an image that is compression-friendly but maintain the performance of machine tasks as high as possible. Our method is simple but effective because additional training is not required by utilizing the pretrained encoder parts of networks for targeted tasks. From the extensive experiments for the object detection and instance segmentation tasks, our method has shown considerable BD-rate improvements compared to the image anchor results of MPEG-VCM reference software. As further works toward video coding for machines, our method can be applied to each frame of a video sequence, and also be improved by taking into account the temporal correlations among frames.

5. ACKNOWLEDGEMENT

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